Pattern Recognition and Machine Learning

Chapter 4: Linear Models for Classification

Reading Notes

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Linear Discriminant Functions

Two-Class Problem:

$$y(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + w_0$$

- Decision boundary: $y(\mathbf{x}) = 0$
- $\mathbf{w}$ is orthogonal to decision surface
- w_0 determines position relative to origin

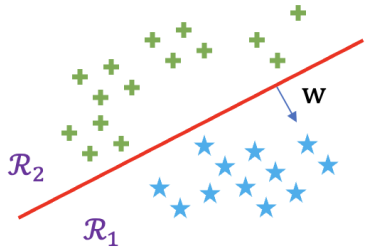
Multiple Classes (K-class discriminant):

- **Per-class** discriminant functions:

$$y_k(\mathbf{x}) = \mathbf{w}_k^\top \mathbf{x} + w_{k0}$$

- Matrix form (with $\mathbf{x}_0 = 1$):

$$\mathbf{y}(\mathbf{x}) = \widetilde{\mathbf{W}}^\top \widetilde{\mathbf{x}}$$



Linear Discriminant for Two Classes

Decision Rule

$k = \arg\max_j y_j(\mathbf{x}) \quad \Leftrightarrow \quad y_k(\mathbf{x}) > y_j(\mathbf{x}) \quad \forall j \neq k$

Least Squares for Classification

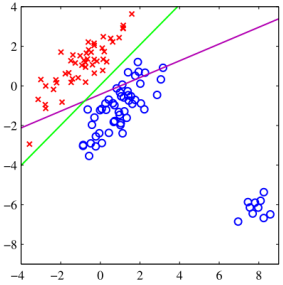
Loss Function:

$$E(\widetilde{\mathbf{W}}) = \frac{1}{2}\mathrm{Tr}\left[(\widetilde{\mathbf{X}}\widetilde{\mathbf{W}} - \mathbf{T})^\top(\widetilde{\mathbf{X}}\widetilde{\mathbf{W}} - \mathbf{T})\right]$$

Optimal Solution:

$$\widetilde{\mathbf{W}} = (\widetilde{\mathbf{X}}^\top\widetilde{\mathbf{X}})^{-1}\widetilde{\mathbf{X}}^\top\mathbf{T}$$

where $\widetilde{\mathbf{X}}$ is the design matrix with bias terms



Advantages:

- Closed-form solution
- Simple implementation
- Fast computation

Limitations:

- Sensitive to outliers
- Lack of probabilistic interpretation
- Poor performance on non-Gaussian data

Key Insight

Least squares minimizes squared error, which is inappropriate for discrete target variables.

Perceptron Algorithm

Perceptron Function:

$$y(\mathbf{x}) = f(\mathbf{w}^\top \mathbf{x})$$

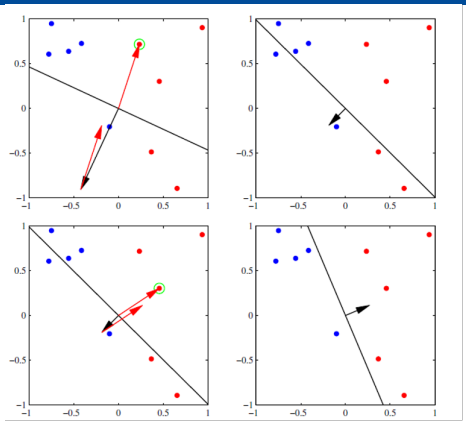
where $f(a) = \begin{cases} +1 & \text{if } a \geq 0 \\ -1 & \text{if } a < 0 \end{cases}$

Error Function:

$$E_P(\mathbf{w}) = -\sum_{n \in \mathcal{M}} \mathbf{w}^\top \mathbf{x}_n t_n$$

Update Rule:

$$\mathbf{w}^{(\tau+1)} = \mathbf{w}^{(\tau)} - \eta \nabla E_P(\mathbf{w}) = \mathbf{w}^{(\tau)} + \eta \mathbf{x}_n t_n$$



Perceptron Update Process

Convergence Theorem

If data is linearly separable, perceptron converges to exact solution in finite steps

Fisher's Linear Discriminant: Intuition

Core Idea: Find a linear projection that maximizes class separation

Key Insight:

- Project data onto 1D line
- Maximize between-class separation
- Minimize within-class variance
- Find optimal projection direction **w**

Projection:

$$y = \mathbf{w}^\top \mathbf{x} \quad (\|\mathbf{w}\| = 1)$$

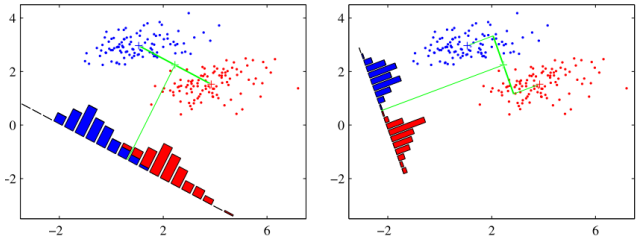


Figure: Fisher's LDA: Maximizing class separation

Geometric Interpretation

Find direction **w** that gives best class separation when data is projected onto this line

Fisher's Linear Discriminant: Mathematics

Within-Class Scatter Matrix:

$$\mathbf{S}_W = \sum_{k=1}^{K} \mathbf{S}_k = \sum_{k=1}^{K} \sum_{n \in \mathcal{C}_k} (\mathbf{x}_n - \mathbf{m}_k)(\mathbf{x}_n - \mathbf{m}_k)^\top \quad \text{where } \mathbf{m}_k = \frac{1}{N_k} \sum_{n \in \mathcal{C}_k} \mathbf{x}_n$$

Between-Class Scatter Matrix:

$$\mathbf{S}_B = \sum_{k=1}^{K} N_k (\mathbf{m}_k - \mathbf{m})(\mathbf{m}_k - \mathbf{m})^\top \quad \text{where } \mathbf{m} = \frac{1}{N} \sum_{n=1}^{N} \mathbf{x}_n$$

Fisher Criterion:

$$J(\mathbf{w}) = \frac{\mathbf{w}^\top \mathbf{S}_B \mathbf{w}}{\mathbf{w}^\top \mathbf{S}_W \mathbf{w}}$$

Optimization Goal

Maximize between-class separation relative to within-class variance

Fisher's Linear Discriminant: Derivation

Optimization Problem:

$$\max_{\mathbf{w}} J(\mathbf{w}) = \frac{\mathbf{w}^\top \mathbf{S}_B \mathbf{w}}{\mathbf{w}^\top \mathbf{S}_W \mathbf{w}}$$

Two-Class Simplification:

$$\mathbf{S}_B = (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^\top$$

Take Derivative:

$$\frac{\partial J(\mathbf{w})}{\partial \mathbf{w}} = 0 \quad \Rightarrow \quad (\mathbf{w}^\top \mathbf{S}_W \mathbf{w})\mathbf{S}_B \mathbf{w} = (\mathbf{w}^\top \mathbf{S}_B \mathbf{w})\mathbf{S}_W \mathbf{w}$$

$$\begin{aligned} \mathbf{S}_B \mathbf{w} &= \lambda \mathbf{S}_W \mathbf{w} \\ (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^\top \mathbf{w} &= \lambda \mathbf{S}_W \mathbf{w} \\ k(\mathbf{m}_1 - \mathbf{m}_2) &= \lambda \mathbf{S}_W \mathbf{w} \\ \mathbf{w} &\propto \mathbf{S}_W^{-1}(\mathbf{m}_1 - \mathbf{m}_2) \end{aligned}$$

Fisher's Linear Discriminant: Classification

After Projection: Data is projected to 1D space

$$y = \mathbf{w}^\top \mathbf{x}, \quad \text{where } \mathbf{w} \propto \mathbf{S}_W^{-1}(\mathbf{m}_1 - \mathbf{m}_2)$$

Projected Class Means:

$$\tilde{m}_1 = \mathbf{w}^\top \mathbf{m}_1, \quad \tilde{m}_2 = \mathbf{w}^\top \mathbf{m}_2$$

Threshold Choices:

- **Midpoint**: $y_0 = \frac{\tilde{m}_1 + \tilde{m}_2}{2}$
- **Sample-weighted**: $y_0 = \frac{N_1 \tilde{m}_1 + N_2 \tilde{m}_2}{N_1 + N_2}$
- **Variance-weighted**: $y_0 = \frac{\tilde{s}_2^2 \tilde{m}_1 + \tilde{s}_1^2 \tilde{m}_2}{\tilde{s}_1^2 + \tilde{s}_2^2}$

Threshold Selection Guide

- **Midpoint**: Equal weighting of both classes
- **Sample-weighted**: Weighted by class sample sizes
- **Variance-weighted**: Weighted by class variances

Generative Approach

Core Idea: Model class distributions, then use Bayes' theorem

Bayes' Theorem (Two-Class):

$$p(\mathcal{C}_1|\mathbf{x}) = \frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1) + p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)}$$

Sigmoid Representation:

$$p(\mathcal{C}_1|\mathbf{x}) = \sigma(a) = \frac{1}{1+\exp(-a)} \quad \text{where } a = \ln \frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)}$$

$$a = \ln\left(\frac{\sigma(a)}{1-\sigma(a)}\right) = \ln\left(\frac{p(\mathcal{C}_1|\mathbf{x})}{p(\mathcal{C}_2|\mathbf{x})}\right)$$

Key Advantage

Can generate synthetic data by sampling from $p(\mathbf{x}|\mathcal{C}_k)$

Continuous Inputs: Gaussian Discriminant Analysis

Model Assumption: Class-conditional distributions are Gaussian with shared covariance

$$p(\mathbf{x}|\mathcal{C}_k) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$$

$$\begin{aligned} a &= \ln \frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)} = \ln \frac{\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_1, \boldsymbol{\Sigma})p(\mathcal{C}_1)}{\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_2, \boldsymbol{\Sigma})p(\mathcal{C}_2)} \\ &= \mathbf{x}^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) - \frac{1}{2}\boldsymbol{\mu}_1^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 + \frac{1}{2}\boldsymbol{\mu}_2^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_2 + \ln \frac{p(\mathcal{C}_1)}{p(\mathcal{C}_2)} \\ \mathbf{w} &= \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \\ w_0 &= -\frac{1}{2}\boldsymbol{\mu}_1^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 + \frac{1}{2}\boldsymbol{\mu}_2^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_2 + \ln \frac{p(\mathcal{C}_1)}{p(\mathcal{C}_2)} \end{aligned}$$

Final Result: $p(\mathcal{C}_1|\mathbf{x}) = \sigma(a) = \sigma(\mathbf{w}^\top \mathbf{x} + w_0)$

Key Insight

Shared covariance structure $\Rightarrow$ Linear decision boundary

Maximum Likelihood Estimation

Likelihood Function:

$$p(\mathcal{D}|\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \boldsymbol{\Sigma}) = \prod_{k=1}^{2} \prod_{n \in \mathcal{C}_k} \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$$

Maximizing with respect to parameters:

- **Class Means**:

$$\frac{\partial}{\partial \boldsymbol{\mu}_k} \ln p(\mathcal{D}|\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \boldsymbol{\Sigma}) = 0 \quad \Rightarrow \quad \boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{n \in \mathcal{C}_k} \mathbf{x}_n$$

- **Shared Covariance**:

$$\frac{\partial}{\partial \boldsymbol{\Sigma}} \ln p(\mathcal{D}|\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \boldsymbol{\Sigma}) = 0 \quad \Rightarrow \quad \boldsymbol{\Sigma} = \frac{1}{N} \sum_{k=1}^{2} \sum_{n \in \mathcal{C}_k} (\mathbf{x}_n - \boldsymbol{\mu}_k)(\mathbf{x}_n - \boldsymbol{\mu}_k)^\top$$

GDA Limitations

Requires shared covariance and Gaussian assumptions, which is sensitive to outliers.

Discrete Inputs: Naive Bayes Classifier

Model Assumption: Features are conditionally independent given class

$$p(\mathbf{x}|\mathcal{C}_k) = \prod_{i=1}^{D} \mu_{ki}^{x_i}(1-\mu_{ki})^{1-x_i} \qquad \mu_{ki} = p(x_i = 1|\mathcal{C}_k)$$

Posterior Derivation:

$$\begin{aligned} p(\mathcal{C}_1|\mathbf{x}) &= \sigma\left(\ln \frac{p(\mathbf{x}|\mathcal{C}_1)p(\mathcal{C}_1)}{p(\mathbf{x}|\mathcal{C}_2)p(\mathcal{C}_2)}\right) \\ &= \sigma\left(\sum_{i=1}^{D} x_i \ln \frac{\mu_{1i}}{\mu_{2i}} + \sum_{i=1}^{D} \ln \frac{1-\mu_{1i}}{1-\mu_{2i}} + \ln \frac{p(\mathcal{C}_1)}{p(\mathcal{C}_2)}\right) \end{aligned}$$

Final Result: $p(\mathcal{C}_1|\mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + w_0)$

Key Insight

Conditional independence assumption $\Rightarrow$ Linear decision boundaries despite discrete inputs

Discriminative vs Generative Approach

Generative Models:

$$p(\mathcal{C}_k|\mathbf{x}) = \frac{p(\mathbf{x}|\mathcal{C}_k)p(\mathcal{C}_k)}{p(\mathbf{x})}$$

Model class-conditional densities

- Learn $p(\mathbf{x}|\mathcal{C}_k)$ and $p(\mathcal{C}_k)$
- Can generate synthetic data
- More parameters needed
- Examples: LDA, QDA, Naive Bayes

Discriminative Models:

$p(\mathcal{C}_k|\mathbf{x})$ modeled directly

(no density modeling)

- Directly learn $p(\mathcal{C}_k|\mathbf{x})$
- Cannot generate data
- Fewer parameters
- Better performance often
- Examples: Logistic Regression

Key Insight

Discriminative models focus only on the decision boundary, ignoring data distribution details

Logistic Regression: Motivation

From Generative to Discriminative:
For two classes with Gaussian distributions having **shared covariance**:

$$p(\mathcal{C}_1|\mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + w_0)$$
$$\mathbf{w} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$$
$$w_0 = -\frac{1}{2}\boldsymbol{\mu}_1^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1 + \frac{1}{2}\boldsymbol{\mu}_2^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_2 + \ln \frac{p(\mathcal{C}_1)}{p(\mathcal{C}_2)}$$

Logistic Regression Approach:

- **Assume**: $p(\mathcal{C}_1|\mathbf{x}) = \sigma(\mathbf{w}^\top \boldsymbol{\phi}(\mathbf{x}))$
- **Estimate**: $\mathbf{w}$ directly from data (MLE/MAP)
- **Advantage**: More flexible, fewer assumptions

Parameter Count Comparison

For D-dimensional features: Generative needs $O(D^2)$ parameters (covariance), Discriminative needs only $O(D)$

Logistic Regression: Model Specification

Binary Classification:

$$p(\mathcal{C}_1|\boldsymbol{\phi}) = \sigma(\mathbf{w}^\top \boldsymbol{\phi}) = \frac{1}{1+\exp(-\mathbf{w}^\top \boldsymbol{\phi})}$$
$$p(\mathcal{C}_2|\boldsymbol{\phi}) = 1 - \sigma(\mathbf{w}^\top \boldsymbol{\phi}) = \sigma(-\mathbf{w}^\top \boldsymbol{\phi})$$

Basis Functions $\boldsymbol{\phi}(\mathbf{x})$:

- Linear: $\boldsymbol{\phi}(\mathbf{x}) = [1, x_1, \ldots, x_D]^\top$
- Polynomial: $\boldsymbol{\phi}(\mathbf{x}) = [1, x_1, x_2, x_1^2, x_1x_2, x_2^2, \ldots]^\top$
- Nonlinear: $\boldsymbol{\phi}(\mathbf{x}) = [1, \phi_1(\mathbf{x}), \ldots, \phi_M(\mathbf{x})]^\top$

Log-Odds Representation:

$$\ln \frac{p(\mathcal{C}_1|\boldsymbol{\phi})}{p(\mathcal{C}_2|\boldsymbol{\phi})} = \mathbf{w}^\top \boldsymbol{\phi}$$

Decision boundary: $\mathbf{w}^\top \boldsymbol{\phi} = 0$ (linear in feature space).

Maximum Likelihood Estimation

Dataset: $\{\phi_n, t_n\}_{n=1}^N$, where $t_n \in \{0, 1\}$ **Model**: $y_n = \sigma(\mathbf{w}^\top \phi_n)$

Likelihood Function:

$$p(\mathbf{t}|\mathbf{w}) = \prod_{n=1}^{N} y_n^{t_n}(1 - y_n)^{1-t_n}$$

Cross-Entropy Error:

$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{w}) = -\sum_{n=1}^{N} [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$

$$\nabla E(\mathbf{w}) = \sum_{n=1}^{N} (y_n - t_n)\phi_n$$

Optimization Characteristics

No closed-form solution; iterative optimization algorithms (gradient descent, Newton's method, etc.) are required.

Iterative Reweighted Least Squares (IRLS)

Newton-Raphson Update:

$$\mathbf{w}^{\text{new}} = \mathbf{w}^{\text{old}} - \mathbf{H}^{-1}\nabla E(\mathbf{w})$$

Hessian Matrix:

$$\mathbf{H} = \nabla\nabla E(\mathbf{w}) = \sum_{n=1}^{N} y_n(1-y_n)\phi_n\phi_n^{\top} = \mathbf{\Phi}^{\top}\mathbf{R}\mathbf{\Phi}$$

where $\mathbf{R}_{nn} = y_n(1-y_n)$.

IRLS Form:

$$\mathbf{w}^{\text{new}} = (\mathbf{\Phi}^{\top}\mathbf{R}\mathbf{\Phi})^{-1}\mathbf{\Phi}^{\top}\mathbf{R}\mathbf{z}$$

where $\mathbf{z} = \mathbf{\Phi}\mathbf{w}^{\text{old}} - \mathbf{R}^{-1}(\mathbf{y}-\mathbf{t})$.

Algorithm Interpretation

Each iteration solves a weighted least squares problem, with weights adjusted according to the current predictions.

Multi-class Logistic Regression: Softmax

Model:

$$p(\mathcal{C}_k|\phi) = \frac{\exp(a_k)}{\sum_{j=1}^{K}\exp(a_j)}, \qquad a_k = \mathbf{w}_k^\top \phi$$

Likelihood Function:

$$p(\mathbf{T}|\mathbf{w}_1,\ldots,\mathbf{w}_K) = \prod_{n=1}^{N}\prod_{k=1}^{K} y_{nk}^{t_{nk}}$$

where $y_{nk} = p(\mathcal{C}_k|\phi_n)$, and $\mathbf{T}$ is the $N \times K$ indicator matrix.

Gradient:

$$\nabla_{\mathbf{w}_j} E = \sum_{n=1}^{N}(y_{nj} - t_{nj})\phi_n$$

Relation to Binary Classification

When $K = 2$, softmax regression reduces to logistic regression (sigmoid function).

Generalized Linear Models

Model: $y = f(\mathbf{w}^\top \phi)$, where f is the activation and f^{-1} the link function

Canonical Link: $f^{-1}(y) = \psi(y)$, the map from the conditional mean to the natural parameter

$$\nabla_{\mathbf{w}} E(\mathbf{w}) = \frac{1}{s} \sum_{n=1}^{N} (y_n - t_n)\phi_n$$

Distribution	$f^{-1}(y)$	$\mathrm{Var}[t]$
Gaussian	y	σ^2
Bernoulli	$\ln \frac{y}{1-y}$	$y(1-y)$
Poisson	$\ln y$	y

$s = \beta^{-1}$ for the Gaussian and $s = 1$ for the logistic; the Bernoulli row is the model of the previous two frames.

Consequence

Without this choice the gradient carries an extra factor $\psi'(y_n) f'(a_n)$, as in the general form.

Laplace Approximation Method

Objective: Approximate complex posterior with Gaussian

Procedure:

1. Find mode $\mathbf{w}_{\text{MAP}}$ of posterior
2. Compute Hessian: $\mathbf{A} = -\nabla\nabla \ln p(\mathbf{w}|\mathcal{D})$
3. Gaussian approximation: $q(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{w}_{\text{MAP}}, \mathbf{A}^{-1})$

Mathematical Basis:

$$\ln p(\mathbf{w}|\mathcal{D}) \approx \ln p(\mathbf{w}_{\text{MAP}}|\mathcal{D}) - \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\text{MAP}})^{\top}\mathbf{A}(\mathbf{w} - \mathbf{w}_{\text{MAP}})$$

Applicability

Works well for unimodal, approximately Gaussian posteriors

Model Evidence and BIC

Model Evidence:

$$p(\mathcal{D}) = \int p(\mathcal{D}|\mathbf{w})p(\mathbf{w})d\mathbf{w}$$

Laplace Approximation:

$$\ln p(\mathcal{D}) \approx \ln p(\mathcal{D}|\mathbf{w}_{\text{MAP}}) + \ln p(\mathbf{w}_{\text{MAP}}) + \frac{M}{2}\ln(2\pi) - \frac{1}{2}\ln|\mathbf{A}|$$

BIC:

$$\ln p(\mathcal{D}) \approx \ln p(\mathcal{D}|\mathbf{w}_{\text{MAP}}) - \frac{M}{2}\ln N$$

AIC:

$$\ln p(\mathcal{D}) \approx \ln p(\mathcal{D}|\mathbf{w}_{\text{MAP}}) - M$$

Model Selection

Trade-off between model fit and complexity

Bayesian Framework

Predictive Distribution:

$$p(\mathcal{C}_1|\mathbf{t}) = \int \sigma(\mathbf{w}^\top \phi) p(\mathbf{w}|\mathbf{t}) d\mathbf{w}$$

Computational Challenges

- Posterior $p(\mathbf{w}|\mathbf{t})$ is non-Gaussian
- Predictive distribution is intractable
- Requires approximation methods

Approximation Methods:

- Laplace approximation
- Variational inference (Chapter 10)
- MCMC sampling (Chapter 11)

Laplace Approximation Application

Posterior Approximation:

$$p(\mathbf{w}|\mathbf{t}) \approx \mathcal{N}(\mathbf{w}|\mathbf{w}_{\text{MAP}}, \mathbf{S}_N)$$

where $\mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \sum_{n=1}^{N} y_n(1-y_n)\boldsymbol{\phi}_n\boldsymbol{\phi}_n^\top$

Predictive Distribution:

$$p(\mathcal{C}_1|\boldsymbol{\phi}, \mathbf{t}) \approx \sigma\left(\kappa(\sigma_a^2)\mu_a\right)$$

where

$$\mu_a = \mathbf{w}_{\text{MAP}}^\top \boldsymbol{\phi}$$
$$\sigma_a^2 = \boldsymbol{\phi}^\top \mathbf{S}_N \boldsymbol{\phi}$$
$$\kappa(\sigma^2) = (1 + \pi\sigma^2/8)^{-1/2}$$

Practical Value

Good balance between computational efficiency and accuracy

Summary

- **Discriminant Functions**: Direct classification approaches
  - Fisher's LDA, perceptron, least squares
- **Generative Models**: Model class conditionals + priors
  - Gaussian assumptions, naive Bayes
  - Can generate synthetic data
- **Discriminative Models**: Direct posterior modeling
  - Logistic regression, IRLS optimization
  - Generally better performance
- **Bayesian Methods**: Full probabilistic treatment
  - Laplace approximation for tractability
  - Uncertainty quantification in predictions

Key Takeaways

- Different assumptions lead to complementary approaches
- Bayesian methods naturally handle uncertainty
- Approximate inference enables practical Bayesian computation